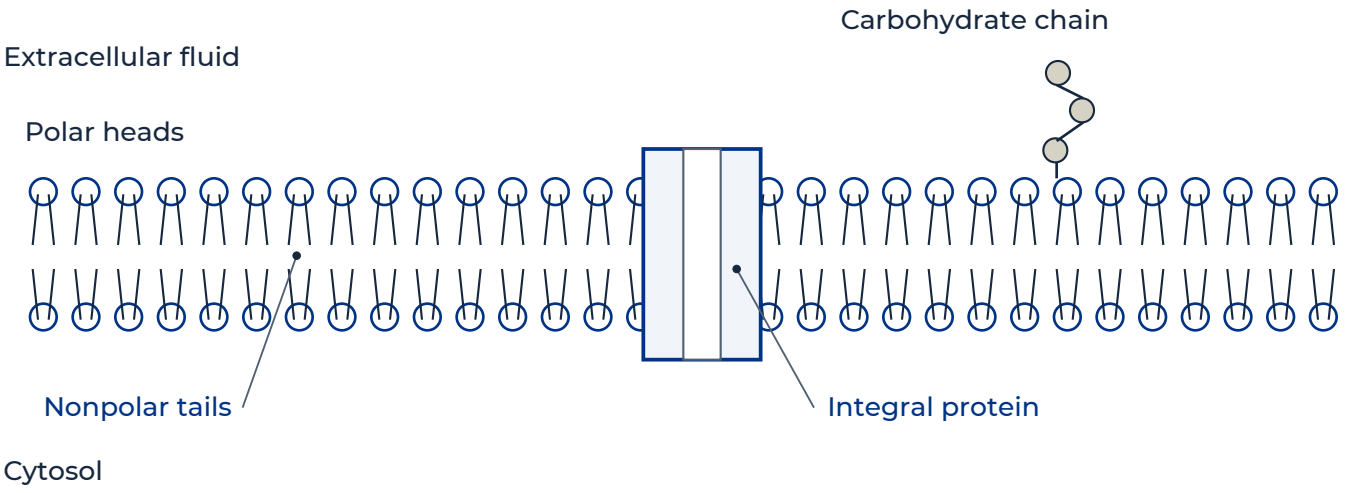
Cell membrane components

The bilayer is a dynamic boundary with selective pathways and unequal faces.



Component	Chemical / functional meaning
Phospholipid	Amphipathic: polar head interacts with water; nonpolar tails face the core. Two leaflets form the bilayer.
Membrane protein	Can transport, receive signals, anchor structures or catalyze reactions. Transmembrane regions fit the hydrophobic core.
Surface carbohydrate	Attached to a lipid or protein on the extracellular face; helps recognition, adhesion or protection.
Fluid membrane	Lipids and many proteins move laterally. It is a dynamic selective boundary, not a solid wall with fixed holes.

Permeability: small nonpolar O2 crosses lipid readily; ions need appropriate pathways. A gradient alone does not guarantee rapid crossing. Cholesterol and many other components are omitted from this schematic.

Cell membrane components

Self-check questions and explanations. Refer to the visual guide on the front.

What are the main parts of this membrane schematic?

Polar phospholipid heads face water; nonpolar hydrocarbon tails form the core; integral proteins span the membrane; extracellular carbohydrate chains attach to lipids or proteins.

Why do phospholipid heads face water while tails face inward?

A phospholipid has one head and two tails in this schematic. Water belongs by the heads on either aqueous side. The tail-rich interior is hydrophobic and excludes most water.

Why is a channel useful for ion movement?

The channel can provide a water-compatible path. A charged ion is energetically disfavored in the nonpolar lipid core; charge and hydration matter, not just size.

How do O2, Na+ and a large protein differ in crossing this membrane?

O2 is small and nonpolar and can diffuse through the lipid region. Na+ may cross via an open selective channel down its electrochemical gradient. A large protein generally needs a specialized route such as vesicular transport.

Which side carries glycoprotein carbohydrate chains?

Carbohydrate chain faces extracellular fluid. It can contribute to cell recognition, adhesion or protection. Do not place the chain on the cytosolic face of the plasma membrane.

Why is a membrane more than a solid wall with fixed holes?

Lipids and many proteins move laterally within a dynamic bilayer. Permeability depends on molecule and available transport machinery. This flat static image omits curvature, motion, asymmetry and molecular diversity.